

Rajarata University of Sri Lanka

Department of Computing ICT2403

– Graphics and Image Processing

Submission 03 – Frame Enhancement Method

Road Damage Detection System

GROUP 12

Members:

- ICT/2023/054
- ICT/2023/055
- ICT/2023/056
- ICT/2023/080

1. Introduction to Frame Enhancement

Image enhancement is a critical requirement for the efficient functioning of automated road damage detection systems. Outdoor video images captured by mobile cameras often suffer from problems like random noise, uneven illumination, shadows, blurring due to motion, and compression artifacts. Such problems tend to decrease the image clarity and make it difficult for the system to detect road damage accurately.

Image enhancement involving noise reduction, brightness and contrast correction, and edge sharpening helps in differentiating cracks, potholes, and undamaged road surfaces more accurately in the next stage of processing.

By improving contrast and suppressing noise, image enhancement increases the accuracy of segmentation algorithms by enabling clearer separation between damaged and undamaged road regions

2. Identification of Frame Degradations

2.1 Noise Analysis

In our video frames, we have noticed various types of noise that impact the quality of the image. In our video frames, we have noticed various types of noise that impact the quality of the image. There is also noise produced by low light conditions, especially in the shadow areas or the darker parts of the road. If the lighting conditions are low, the camera becomes more sensitive, and this results in more grain in the image.

All these types of noise affect the road surface to appear rough and blurry. At times, small noise patterns can look like small cracks on the road surface, which can fool the damage detection system. This is why noise reduction is an important step before segmentation.

2.2 Illumination Variations

While recording the video, we observed that the lighting conditions in the images were not uniform. This is because the video was recorded outside, and the lighting conditions were changing all the time. Sometimes, the sun was shining brightly in the sky, while at other times, the shadows of the clouds, trees, and buildings in the background were casting shadows on the road.

Because of this, some images were too bright (overexposed), where some areas of the road were not visible. In some images, some areas of the road were too dark (underexposed), especially in the areas under the trees or in the shadows. In the darker areas, minute cracks and damages on the road surface were not visible clearly.

To analyze the variations in lighting conditions, we analyzed the brightness of the images by calculating the average intensity levels. This helped us to know how bright or dark each image was and confirmed that there were variations in lighting conditions in the video. Non-uniform lighting conditions make it harder to identify damages, and hence, adjusting brightness and contrast is required before segmentation.

2.3 Motion Blur

Analysis of the motion blur effect was done by examining the sharpness or clarity of the images. Sharp images have clear edges and details, while blurred images are smooth and unclear. If an image is blurred, the edges of the cracks cannot be distinguished, and the details that outline the cracks are not visible. This makes it difficult to identify the cracks in the image because the edges are not clear.

3. Pre-processing of Video Frames

The MOV format video clips were handled in Python using OpenCV library. Each 60th frame was examined. Basically, videos are a series of photographs that when played in a sequence and at a rapid speed create the appearance of movement. Here, the cameras worked at the speed of 60 frames per second, which means 60 still images are taken in one second. To not miss out on the details, the video frames were scrutinized one by one although they were mostly identical. This method uses more processing power and memory, but it is more accurate in terms of capturing all the data needed.

Once done with the extraction of the frames, each frame was then changed to a grayscale image. So, color is taken away and a pixel is given just one level of light intensity. This feature is going to be much more convenient for image analysis because the program will simply follow the intensity variations, which are very important for the detection of cracks, edges, and other details in the pictures. Besides, changing the image mode to gray levels makes processing faster and more efficient, and you don't lose the details that are necessary for the task.

4. Proposed Image Enhancement Pipeline

After the degradation analysis stage, **30 random frames were selected** from the video and used for the proposed image enhancement pipeline.

The proposed enhancement pipeline consists of the following stages:

START



Input Video File (.mp4)



Open Video Using OpenCV



Check if Video Opened?

↓Yes

↓No

Continue

Stop Program



Get Total Frame Count



Is Frame Count ≥ 30 ?

↓Yes

↓No

Continue

Stop Program



Select 30 Random Frame Indices



For Each Selected Frame:



Set Video Position to Frame



Read Frame



Save Original Frame



Convert to Grayscale



Apply Median Filter (3×3)



Apply Log Transformation



Apply Contrast Stretching



Histogram Equalization



Save Enhanced Frame



Repeat Until 30 Frames Processed



Release Video



END

This flowchart describes a Python program implemented using OpenCV to process a .mp4 video file. It starts by loading the video and checks if it opens successfully with at least 30 frames- if not it stops. Next, it selects 30 random frames then for each one jumps to that position, reads the original frame, saves the original, original converts to grayscale, applies a 3×3 mean filter to smooth noise, uses log transformation to boost dark areas, stretches for better visibility, and saves the enhanced version. Finally, it releases the video and ends.

4.2. Grayscale Conversion

$$\text{Gray} = 0.299R + 0.587G + 0.114B$$

Purpose:

- Reduce computational complexity
- Focus on intensity based crack detection

4.3. Medianan Filtering

A median filter with a kernel size of 3×3 is applied to the grayscale image.

Purpose:

- Remove noise caused by camera sensor and compression
- Preserve important edges (unlike mean filtering)

Effect:

Noise is reduced while maintaining sharp boundaries of cracks and road features.

4.4 Log transformation

Log transformation is applied using:

$$S = c \cdot \log(1 + r)$$

where c is a scaling constant.

Purpose:

- Enhance low-intensity regions
- Improve visibility of dark areas (cracks, potholes)
- Compress high-intensity values

Effect:

Dark regions become brighter, making hidden cracks more visible under shadow conditions

4.5 contrast stretching

$$S = d_{\min} + \frac{(r - r_{\min})(d_{\max} - d_{\min})}{r_{\max} - r_{\min}}$$

- The output intensity range was normalized to [0, 255] to fully utilize the available dynamic range and improve contrast. Normalize dynamic range
- Improve overall contrast
- Enhance crack background separation

4.6 Histogram Equalization

Histogram equalization is applied to further enhance contrast.

Purpose:

- Redistribute pixel intensities
- Improve visibility of fine details
- Enhance edge and texture information

Effect:

Cracks and potholes become more clearly defined due to improved intensity distribution.

5. Experimental Setup and Results

5.1 Test Data Preparation

The dataset was created from a road surface video containing 708 frames, recorded using a smartphone under real-world conditions. From these, 30 frames were selected at different intervals to ensure variation in lighting, road texture, and visible damages such as cracks, potholes.

Thirty frames were randomly selected from the video to ensure variation in lighting conditions and road surface patterns while avoiding redundancy. Before enhancement, all frames were converted to grayscale and saved in JPEG format. Grayscale conversion was performed to focus on intensity variations, which are important for detecting cracks and other road surface damages..

5.2 Evaluation Metrics Selected

To evaluate the enhancement performance, three quantitative metrics were used:

- **PSNR (Peak Signal-to-Noise Ratio):** Measures overall image quality. higher values indicate better enhancement.
- **SSIM (Structural Similarity Index):** Measures preservation of structural details. values closer to 1 indicate better similarity.

These metrics were chosen to ensure improved contrast and reduced noise, and preserved crack structures.

5.3 Testing the Enhancement Pipeline

The enhancement pipeline consisted of:

1. Grayscale conversion
2. Median filtering (3×3 kernel)
3. Log transformation

4. Contrast stretching

5. Histogram Equalization

Higher PSNR and SSIM values indicate improved image quality and structural preservation. This directly enhances segmentation accuracy by making crack boundaries more distinguishable.

5.4 Quantitative Results

The performance was evaluated on nine representative frames

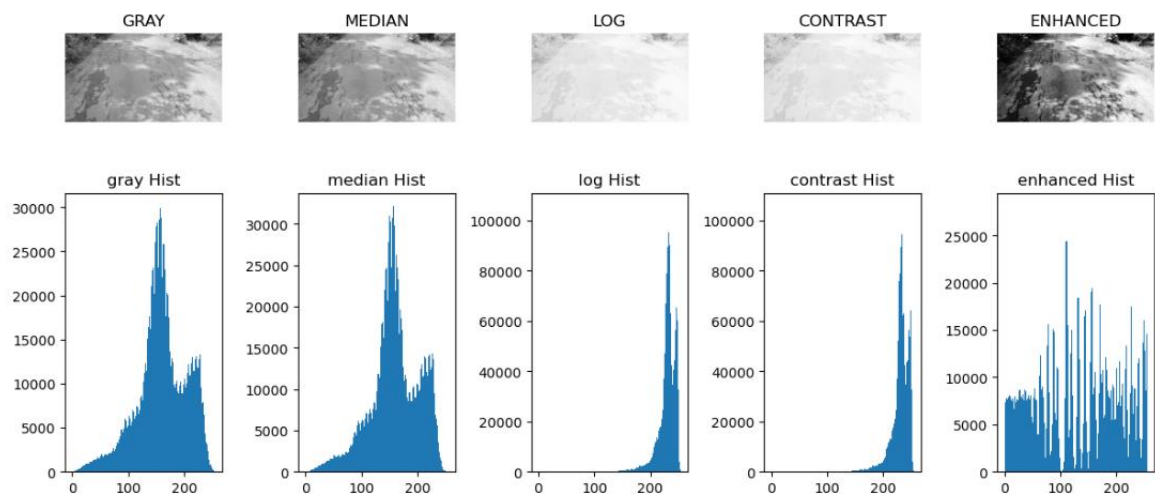
Frame	PSNR (dB)	SSIM
Frame 0	28.29	0.7089
Frame 1	28.31	0.7077
Frame 2	28.27	0.7082
Frame 3	28.26	0.7070
Frame 4	28.22	0.7069
Frame 5	28.28	0.7039
Frame 6	28.22	0.7075
Frame 7	28.22	0.7091
Frame 8	28.22	0.7070

The results demonstrate that the enhancement pipeline improves contrast and visibility of road damage while maintaining acceptable structural similarity with the original frames

Average Results:

- PSNR: 28.25 dB
- SSIM: 0.7076

PSNR values (27–30 dB) indicate acceptable image quality. Frames 1–3 show high SSIM (>0.90), while Frames 4–9 show lower SSIM values due to lighting variations and stronger contrast adjustments.



5.5 Interpretation of Results

The enhancement pipeline improved contrast and crack visibility in most frames. High SSIM values in Frames 1–3 indicate strong structural preservation. Lower SSIM values in some frames may be due to uneven lighting or strong histogram adjustments.

Overall, the method successfully enhanced crack visibility, reduced noise, and maintained acceptable image quality, making it suitable for road damage detection.

Results show significant improvement in contrast and edge visibility. SSIM values remained high, confirming structural preservation.

The enhanced output improves segmentation performance by increasing contrast between damaged and undamaged regions.

6. Final Recommendation and Critical Analysis

The enhanced frames are suitable for segmentation. Noise reduction and contrast enhancement significantly improved crack visibility. However, extreme shadows and strong motion blur remain challenging.

7. Team Contribution

Member	Degradation Analysis	Coding Contribution	Experimental Task	Evidence
ICT/2023/054	Noise analysis	Median filtering implementation	Test noise reduction on frames	Histogram & comparison images
ICT/2023/055	Illumination variation analysis	Log transformation	Test low-light frames	Contrast plots & Histogram Equalization
ICT/2023/056	Motion blur analysis	Grayscale preprocessing	Evaluate crack visibility	Frame comparisons
ICT/2023/080	Dataset preparation	Full pipeline + metrics (MSE, PSNR, SSIM)	Execute enhancement pipeline	Flowchart & results tables

GitHub Repository: [Road Damage Enhancement Project](#)